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An electrically conductive gellan gum/polyvinyl alcohol interpenetrating network hydrogel: a dual crosslinked 3D printing ink for cardiac tissue

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Biofabrication of cardiac tissue constructs with inherent electrical conductivity and contractility presents a significant challenge. In this study, an interpenetrating network (IPN) hydrogel composed of methacrylate-modified polyvinyl alcohol (M-PVA) and gellan gum (GG) reinforced with reduced graphene oxide (rGO) has been developed. The M-PVA/GG/rGO hydrogel leverages the thermoresponsive property of polysaccharide gellan gum for controlled gelation during the 3D printing process, followed by post-printing photocrosslinking of M-PVA to enhance structural stability. The IPN hydrogel exhibited porous morphology with interconnected pores, high porosity, swellability, and significant electrical conductivity ($0.62 \pm 0.05 \text{ mS cm}^{-1}$) imparted by the inclusion of rGO. Rheological analysis demonstrated the shear-thinning property and predominant elastic modulus of the developed hydrogel, thereby being suitable for pneumatic extrusion-based 3D printing. The printed constructs cultured with H9c2 cardiomyoblasts and EA.hy926 endothelial cells demonstrated favorable *in vitro* cell viability, proliferation, and cardiac specific gene expression, influenced by the matrix composition. The dual-crosslinked, electroconductive M-PVA/GG/rGO hydrogel shows significant promise for promoting vascularization in cardiac tissue engineering, facilitating tissue regeneration, development of organotypic models and potentially enabling the development of electroconductive biomedical devices.

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1. Introduction

Myocardial infarction (MI), resulting from occlusion of coronary arteries and consequent deprivation of oxygen and nutrients to the myocardium, remains a leading cause of morbidity and mortality worldwide.¹ The ensuing cardiomyocyte death initiates extracellular matrix (ECM) degradation and pathological remodelling processes, including scar formation, ventricular wall thinning, and chamber dilation, ultimately progressing to heart failure. A key challenge in MI repair is the disruption of electrical conductivity caused by the replacement of the native myocardium with fibrotic scar tissue. This severely impairs impulse propagation and cardiac performance.^{2,3} Restoration of electromechanical coupling is critical to resynchronize myocardial contractions and prevent adverse remodeling.⁴ Current clinical treatments mainly address symptom management

without promoting tissue regeneration, underscoring the need for regenerative strategies that restore myocardial function.^{5,6} To address these gaps, fabrication of cardiac tissue-mimetic constructs using hydrogel-based systems, incorporating electroactive conducting nanomaterials and therapeutic cells, has emerged as a promising strategy. For instance, Jing *et al.* developed a polydopamine-incorporated chitosan/GO hydrogel with self-adhesive, self-healing, and electrically conductive properties (1.22 mS cm^{-1}). This hydrogel enhanced the viability and proliferation of human embryonic stem cell-derived fibroblasts (HEF1) and cardiomyocytes (CMs) compared to GO-free hydrogels.⁷ Similarly, Tianxiao *et al.* fabricated a 3D-printed MSC@GO hydrogel patch composed of fibrinogen scaffolds, mesenchymal stem cells (MSCs), and graphene oxide for cardiac repair after MI. The patch provided electrical integration and mechanical support and promoted regeneration. MSC@GO reduced cardiomyocyte apoptosis, increased connexin 43 (Cx43) expression, and improved cardiac function in a mouse MI model.⁸ These constructs replicate the micro- and nanoscale architecture, as well as the mechanical and electrical properties of the native myocardium, while modulating the post-infarction microenvironment to promote myocardial regeneration and function restoration.

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Nanomaterials with electrical conductivity like reduced graphene oxide (rGO),^{9,10} carbon nanotubes^{11,12} and gold nanostructures^{13,14} show great promise in enhancing electrical signal propagation in damaged cardiac tissue. GO's excellent processability and cell-interactive properties make it suitable for integration into hydrogels, but its low electrical conductivity limits its electromechanical applications.¹⁵ rGO, produced by partially restoring GO's sp²-carbon network, improves conductivity while retaining bioactive functional groups. Its biocompatibility, ease of dispersion, and superior surface functionalization make rGO preferable over carbon nanotubes (CNTs), which are more difficult to process and less hydrophilic.¹⁶ Incorporating rGO into non-conductive hydrogels leads to the formation of conductive networks that mimic the electrical behavior of the native myocardium, promoting synchronized cardiomyocyte contractions. Shin *et al.* demonstrated that an rGO-incorporated GelMA hydrogel improved the conductivity and mechanical properties of cardiac scaffolds, enhancing tissue regeneration post-MI.⁹ Similarly, Zhao *et al.* (2022) showed that rGO functionalized electrospun silk cardiac patches improved electrical signal propagation, angiogenesis and functional recovery post-MI, highlighting rGO's potential in restoring electromechanical coupling and improving heart function.¹⁷

3D bioprinting is an advanced technology that enables the fabrication of complex, biomimetic tissue constructs by spatially organizing cellular and acellular components in three dimensions.^{18,19} The bioprinting process consists of three main stages: pre-printing, printing, and post-printing.²⁰ Pre-printing involves material formulation and parameter optimization, printing ensures spatial precision, and post-printing supports printed construct stabilization.²¹ A critical aspect across these stages is the crosslinking mechanism, which can be physical,^{22,23} chemical, or enzymatic^{24,25} and governs material solidification and structural fidelity. Pre-printing crosslinking can be employed to enhance viscosity and printability, but excessive crosslinking hinders extrusion and reduces architectural control. Post-printing crosslinking enables precise tuning of mechanical properties and porosity. Overall, a strategic combination of pre-printing and post-printing crosslinking is often adopted to optimize the printability, micro-structure and long-term stability of printed constructs.^{26,27} The ideal ink for cardiac tissue should have optimal mechanical characteristics, electrical conductivity and a favorable environment for cell survival, proliferation, and differentiation.

Gellan gum (GG), a thermoresponsive polysaccharide derived from *Pseudomonas elodea*, is widely used in tissue engineering owing to its biocompatibility, biodegradability, and ability to support cell adhesion.^{28,29} Upon cooling, GG undergoes physical gelation through helix aggregation, making it suitable for gentle cell encapsulation in pre-printing stages of 3D bioprinting.³⁰ However, its mechanical strength is insufficient for contractility of cardiac tissue constructs. Zargar *et al.* explored rGO-incorporated GG hydrogels for cardiac tissue engineering, noting that while GG supports cell adhesion and tissue regeneration, its mechanical properties require enhancement for long-term cardiac repair.³¹ To address this limitation,

photo-crosslinkable polyvinyl alcohol methacrylate (M-PVA) has been chosen in this study. M-PVA enables covalent crosslinking *via* photo-initiated polymerization, enhancing the mechanical strength and stability of bioprinted constructs while maintaining cell compatibility. Thus, integrating thermoresponsive GG for pre-printing gelation and M-PVA for post-printing crosslinking aims to fabricate robust, biologically active cardiac tissue constructs.³² This dual strategy ensures cell encapsulation and provides optimal mechanical properties to withstand the dynamic forces of cardiac tissue, facilitating long-term tissue regeneration.^{33–35}

In this study, an electro-conductive composite hydrogel based printable ink has been developed by blending natural polysaccharide gellan gum with synthetic M-PVA and incorporating rGO for the fabrication of cardiac tissue-mimetic constructs. Thermoresponsive gelation of gellan gum was integrated with photocrosslinking of M-PVA to form an interpenetrating network (IPN) hydrogel system (Scheme 1). The incorporation of rGO reinforces the viscoelastic property and imparts electro-conductivity, enabling the composite to impart electrical behaviour to 3D printed constructs. This hybrid hydrogel system serves as an optimal ink for pneumatic-based 3D printing, enhances the mechanical properties of the scaffold, promotes cell adhesion, supports cellular interaction and enhances cardiac specific gene expression. The printing parameters and high-resolution 3D patterns of developed conductive polymer composite IPN hydrogels were optimized. The developed 3D printed constructs were characterized for viscoelasticity and electroconductivity. The biocompatibility and neovascularization potential were assessed *in vitro* using rat cardiac myoblast cells (H9C2) and endothelial cells (EA cell line) respectively, demonstrating the composite's potential in cardiac tissue engineering applications.

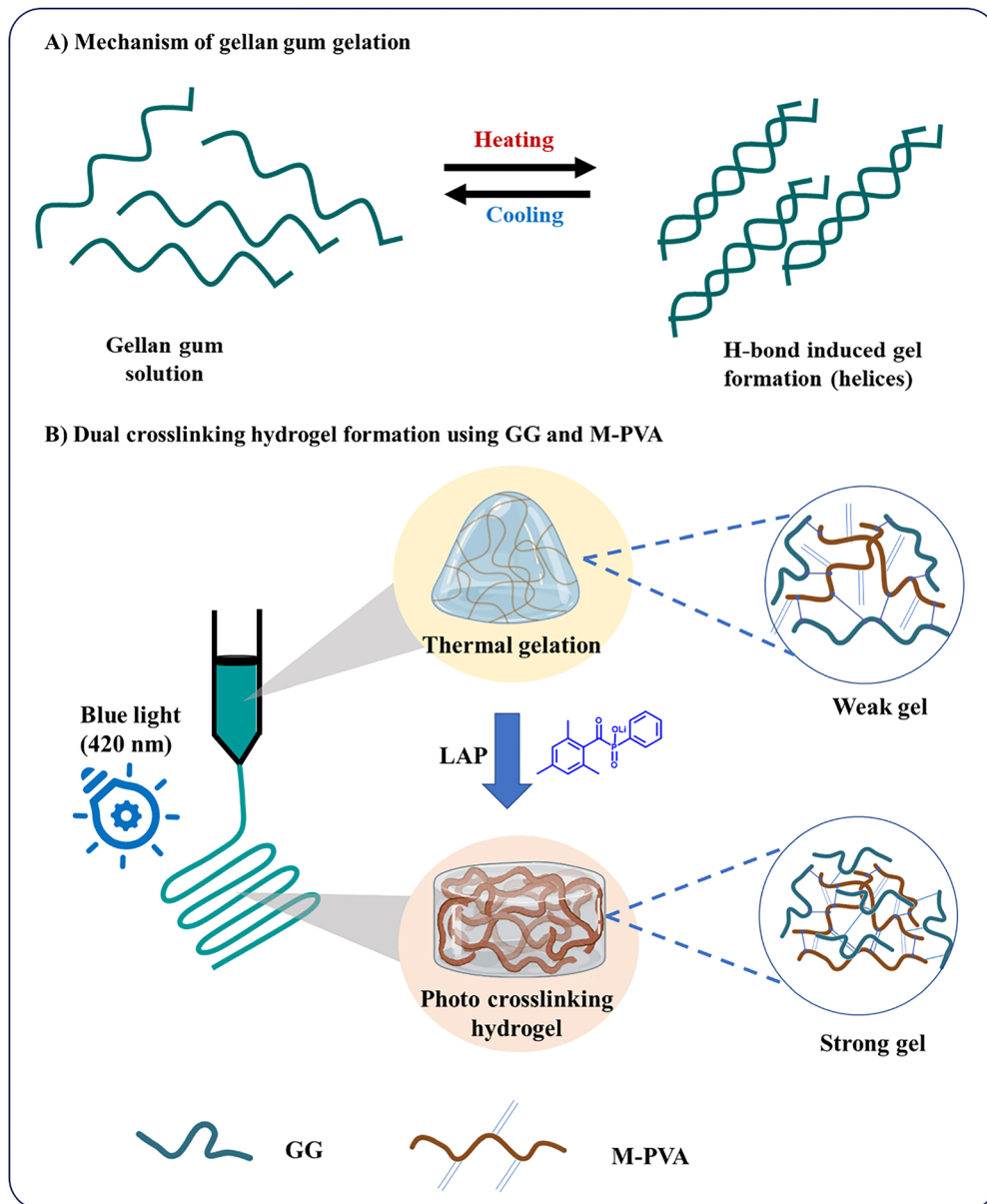
2. Materials and methods

2.1. Materials

Polyvinyl alcohol (PVA) – MW 65 000–125 000, gellan gum, dimethyl amino pyridine (DMAP), dimethyl sulfoxide (DMSO), potassium permanganate (KMnO₄), hydrogen peroxide (H₂O₂), and the WST reagent kit were purchased from HiMedia, India. Glycidyl Methacrylate (GMA) was purchased from SRL, India. Hydrazine hydrate, lithium phenyl-2,4,6-trimethylbenzoylphosphinate (LAP), Dulbecco's modified Eagle's medium (DMEM), penicillin–streptomycin solution and trypsin-EDTA, phalloidin 565, DAPI, fluorescein diacetate (FDA) and propidium iodide (PI) were purchased from Sigma-Aldrich, India.

2.2. Synthesis of reduced graphene oxide (rGO)

Graphene oxide (GO) was prepared from graphite according to modified Hummers' method.³⁶ Briefly, 1g of graphite was added to 23 mL of sulphuric acid and stirred under cool conditions (ice bath) for 1 hour. Sodium nitrate was added dropwise and the stirring continued for additional 2 hours. Subsequently, 3 g of KMnO₄ was added, and the reaction



Scheme 1 Schematic representation of the crosslinking mechanisms using gellan gum and methacrylated polyvinyl alcohol that forms an interpenetrating network (IPN) hydrogel.

was allowed to cool and left overnight at 27 to 30 °C. The reaction was terminated using 92 mL of distilled water and 5 mL of H₂O₂. The resulting mixture was centrifuged, rinsed several times with alcohol and water, and dried in a hot air oven at 90 °C for 24 hours. For reduction of GO, 100 mg of GO was dispersed in distilled water by ultrasonication. Hydrazine hydrate (1.6 mL) was added as a reducing agent, and the reaction mixture was heated at 80 °C for 24 hours. The reaction was quenched by adding 6 mL of 6N HCl, followed by multiple washes with ethanol and water. The product was centrifuged and dried in a hot air oven at 60 °C overnight.

2.2.1. Physicochemical characterization of rGO. The absorbances of rGO and GO were characterized using a UV-visible

spectrometer (Jasco V-750, Japan). Aqueous suspensions of GO and rGO were scanned in a wavelength range of 200–800 nm.

To describe the interlayer spacing, X-ray diffraction patterns of GO and rGO were analysed using a powder X-ray diffractometer (Rigaku) at a scanning rate of 1° per minute in a 2θ range from 5° to 80° with Cu-K α radiation (λ 1.5418 Å).

The structural changes of GO and rGO have been determined effectively using Raman spectroscopy. Raman spectra were recorded for GO and rGO between 500 and 3000 cm⁻¹ on a Ram Aramis Raman spectrometer, using a 514.5 nm Ar laser at 0.5 mW power. The laser spot size was around 1 mm.³⁷

A field emission scanning electron microscope (FE-SEM), CLARA (TESCAN, Europe), operating at an accelerating voltage of 10 keV was used to examine the surface morphology of

synthesised GO and rGO. Using a plasma gold coater, GO and rGO were sputter-coated in a vacuum for 120 seconds prior to examination under a SEM. Synthesized rGO was suspended in water (0.5 mg mL^{-1}), sonicated to obtain homogeneous suspension, and coated on a copper grid for imaging under a transmission electron microscope (Tecnai G2 T20) to determine the nano-sheets of rGO.

2.3. Methacrylation of PVA (M-PVA) and characterization

PVA was methacrylated using the previous procedure with slight modifications.³⁸ Briefly, to 5% w/v PVA solution in DMSO, 2 g of DMAP was added under a nitrogen atmosphere for 1 hour. To it, 0.3 g of GMA was added, and the mixture was stirred for 48 hours at 37°C under dark conditions. The solution was dialyzed extensively against water to remove any unreacted GMA. The dialyzed M-PVA solution was then freeze-dried and stored in an amber tube until further use.

2.3.1. Physicochemical characterization of M-PVA. Fourier transform infrared (FTIR) spectra of PVA and M-PVA were acquired using the JASCO FTIR4700 spectrometer. 10 mg of PVA and M-PVA were mixed with 200 mg of potassium bromide to make a pellet using a casting method. The spectra were recorded at a resolution of 8 cm^{-1} and 32 scans of the spectrum were accumulated.

The chemical structure of M-PVA was determined by NMR analysis. An Avance III HD instrument equipped with a 400 MHz narrow-bore FT-NMR probe (Bruker, Switzerland) was used to record the ^1H NMR spectra of 10 mg PVA and M-PVA dissolved in 0.5 mL of D_2O . The degree of substitution (DS) was calculated from the ^1H NMR spectrum, using the following equation:³⁹

$$\text{DS (\%)} = \frac{A(\text{H}') + A(\text{H}'')/2}{A(\text{H}') + A(\text{H}'')/2 + A(\text{OH})} \times 100 \quad (1)$$

In the ^1H NMR spectrum, $A(\text{H}')$ and $A(\text{H}'')$ represent the areas corresponding to two-vinyl hydrogen from M-PVA and $A(\text{OH})$ represents the area corresponding to hydroxyl hydrogen present in PVA.

2.4. Photo-crosslinkable hydrogel formation and characterization

To fabricate a photocrosslinkable hydrogel, varying concentrations of M-PVA and a photo-initiator (LAP) in PBS were used to assess the hydrogel formation (Table 1). Polymeric solution was irradiated by blue light (420 nm) exposure for 1 to 5 minutes and the gelation time assessed by a conventional tube tilting method. Subsequently, dual-crosslinked hydrogels were fabricated by adding the thermoresponsive polymer gellan gum and rGO into an optimized solution of 5% w/v M-PVA in PBS and 0.15% w/v of LAP (Tables S1 and S2). Polymeric precursor solution was allowed to undergo thermal gelation, followed by 2 minutes of blue light irradiation ($\lambda = 420 \text{ nm}$) to induce photocrosslinking of M-PVA and form interpenetrating network (IPN) hydrogels.

2.4.1. Physico-chemical characterization

FTIR spectroscopy. The FTIR spectra of the lyophilized hydrogel were recorded using a JASCO FTIR4700 spectrometer at a

Table 1 Optimization of photocrosslinkable hydrogel formation using M-PVA

S. no.	M-PVA (% w/v)	LAP (% w/v)	Blue light (420 nm wavelength)	Time (minutes)	Observation
1	1	0.15	420	5	Incomplete gel
2	2	0.15	420	3	Weak gel
3	3	0.15	420	2	Weak gel
4	4	0.15	420	2	Stable gel
5	5	0.15	420	1	Stable gel

resolution of 8 cm^{-1} ; 32 scans of the spectrum were accumulated.

Differential scanning calorimetry (DSC). The glass transition temperature (T_g) and melting temperature (T_m) of lyophilized hydrogels were determined using a DSC PerkinElmer Q200 (TA Instruments, USA). Thermal behaviour was recorded in a nitrogen environment at a rate of 10°C per minute; 5–10 mg of lyophilized hydrogels was weighed and heated twice, from 37°C to 150°C and 37°C to 400°C .

FE-SEM analysis. FE-SEM (CLARA TESCAN, Europe) operating at an accelerating voltage of 10 keV was used to examine the surface morphology of lyophilized hydrogels, sputter-coated prior to examination under a SEM. Images were analyzed using Image J software, and the pore diameters were measured.

Porosity. The porosity of the hydrogel scaffolds was measured using a specific gravity bottle (Hubbard, Hanil, Korea) based on Archimedes' principle and was calculated using the following formula:

$$\text{Porosity (\%)} = \frac{(W_2 - W_3 - W_s)}{(W_1 - W_2)} \times 100 \quad (2)$$

where W_1 is the weight of specific gravity bottle filled with ethanol, W_2 is the weight of specific gravity bottle with ethanol and lyophilized hydrogel submerged, W_3 is the weight of specific gravity bottle with ethanol after removal of hydrogel scaffold and W_s is the initial scaffold weight.

Swelling behavior. Pre-weighed lyophilized hydrogels (W_d) were submerged in PBS at 37°C for 1, 2, 3, 6, 12, and 24 hours to determine the equilibrium swelling state and the equilibrium swelling ratio. At regular intervals, the swollen hydrogels were taken out and wiped off of excess water using filter paper. A weighing balance (Shimadzu, Japan) was then used to determine the weight of the swollen hydrogel (W_s). Each measurement was taken in triplicates, and the swelling ratio was calculated using the formula:

$$\text{Swelling ratio} = \frac{(W_s - W_d)}{W_d} \quad (3)$$

where W_s represents the weight of the swollen hydrogel and W_d represents the initial weight of the lyophilized hydrogel.

In vitro degradation. Pre-weighed lyophilized hydrogels were immersed in PBS at 37°C and 50 rpm. PBS was replenished three days a week and hydrogels were collected on days 1, 3, 5, 10, 15, 20 and 25. Hydrogels were wiped off of excess water using filter paper, freeze-dried and weighed. The following

formula was used to determine the percentage of weight reduction:

$$\text{Weight loss (\%)} = \left[\frac{W_0 - W_t}{W_0} \right] \times 100 \quad (4)$$

where W_0 represents the initial weight of the hydrogel and W_t corresponds to degraded weights of hydrogels at the stipulated time points.

2.4.2. Electrical conductivity. The current–voltage (I – V) measurements were performed following the method of Selvakumar R. *et al.*, using four-probe apparatus with a Keysight B2911A source meter (California, USA).⁴⁰ The bias voltage over the range of -3 V to $+3$ V was applied to the hydrogel of a fixed length of 2 cm. Dual crosslinked hydrogels of 2×2 cm area and 3 mm thickness were prepared on glass slides, and four electrodes were placed on four sides of the hydrogel for measurement using a four-probe setup. The electrical conductivity was measured using the formula:

$$\sigma = 1/\rho \quad (5)$$

where σ represents the electrical conductivity and ρ corresponds to the resistivity of the hydrogel.

Additionally, an LED circuit powered by a 9 V battery was used to assess the electrical conductivity of the hydrogel.

2.4.3. Rheology. Rheological measurements of M-PVA/GG and M-PVA/GG/rGO based dual-crosslinked hydrogel was performed using a Kinexus ultra+ rheometer (Nexus Analytics, Singapore) to analyse the viscoelasticity and dynamic mechanical property. An anti-slip parallel plate geometry was used for hydrogels of 20 mm diameter and 1 mm thickness. The linear viscoelastic region (LVER) was determined by an amplitude sweep test, by varying the strain (γ) from 0.1% to 100% at a constant frequency ($f = 1$ Hz). Subsequently, frequency sweep tests were performed within the LVER at a fixed strain of $\gamma = 0.1\%$, covering an angular frequency (ω) range of 0.1–100 rad s^{-1} with a consistent gap size of 1 mm.⁴¹ The shear thinning behavior of the hydrogel precursor was studied by varying the shear rate from 1 to 1000 s^{-1} logarithmically and the flow behavior of the hydrogels was modelled using the power law equation (Herschel–Bulkley model):

$$\tau = \tau_0 + K\dot{\gamma}^n \quad (6)$$

where τ is the shear stress (Pa), τ_0 is the yield stress (Pa), $\dot{\gamma}$ is the shear rate, K is the consistency index, and n is the flow behavior index.⁴² To study the shear recovery of M-PVA/GG and M-PVA/GG/rGO, a three interval thixotropy analysis was performed. This involved an oscillating time sweep at a constant frequency of 1 rad s^{-1} and a shear strain of 1%. All experiments were performed at 25 °C.

2.5. Optimization of the 3D printing process, pre-printing and post-printing crosslinking in the hydrogel

The interpenetrating polymer network (IPN) was developed using GG at concentrations ranging from 0.5% to 2% and optimal concentration of M-PVA as precursor solution. Blending with thermoresponsive GG facilitated hydrogel formation

upon subjecting to low temperature. This provided network stabilization of hydrogels during the pre-printing process. Electrical conductivity was incorporated into the hydrogel, by introducing rGO (0.15 mg mL^{-1}) into polymeric precursor solution that eventually forms a hydrogel network by integrating thermoresponsive pre-printing and photo-crosslinkable post-printing processes. A pneumatic extrusion-based 3D bio-printer (NBIL-TRIVIMA, Bangalore, India) was used to investigate the 3D printability of the M-PVA/GG hydrogel. Different 3D designs (square, star, heart, and circles) were created using CAD software, and the precursor hydrogel was loaded into 3 mL syringes with a 22-gauge size needle for printing. Pneumatics based extrusion was performed at a feed rate of 600 mm s^{-1} with an applied pressure of 0.8–1 bar to evaluate hydrogel shape retention in a 3D grid. 3D printing was carried out with a nozzle temperature of 25 °C and a relative humidity maintained between 30 and 40%. After printing and 2-minute light exposure, the printed constructs were compared to initial CAD models. The characterization of line width and thickness of the printed grid structure was performed using Image J software.

2.6. In vitro evaluation

2.6.1. Cell culture and maintenance. The *in vitro* cytocompatibility of developed electrically conductive 3D printed constructs for cardiac tissue engineering was evaluated using the cardiac ventricular myogenic cell line (H9C2) and the endothelial cell line (EA). The cells were cultured in DMEM high glucose supplemented with 10% (v/v) fetal bovine serum and 1% penicillin – streptomycin. Polymeric precursor solution for 3D printing was prepared in a sterile environment. In a 12-well plate, the thermo-responsive hydrogel was 3D printed with dimensions of 2×2 cm and exposed to blue light to initiate photocross-linking, followed by culturing of 1×10^4 cells per well on the printed construct for further assays.

2.6.2. Cell viability and metabolic assay. The metabolic activity of cells cultured on developed 3D printed constructs was assessed by WST (water-soluble tetrazolium salt) assay. WST assay quantifies the enzymatic conversion of a water-soluble tetrazolium salt into formazan dye by metabolically active cells. After pre-determined time intervals of 1, 2, and 3 days for EA hy926 cells and 1, 3, 5 and 7 days for H9C2 cells, the culture medium was removed and replaced with the medium containing 10% v/v of WST. The cells were incubated for 3 to 4 hours at 37 °C with 5% CO_2 , and absorbance was measured at 450 nm using an Epoch2 microplate reader. Cell viability upon culturing in the printed construct was assessed using fluorescence-based imaging of live/dead staining with fluorescein diacetate (green, Ex/Em = 495/515 nm) and propidium iodide (red, Ex/Em = 535/615 nm) respectively. Images of the stained cells were captured using a fluorescent microscope (Leica Microsystems DMi8, Germany).

2.6.3. Cytoskeleton staining. Cardiomyoblast cell line of rat origin (H9C2) enriched printed constructs were fixed for one hour at 37 °C using 4% paraformaldehyde and washed with phosphate-buffered saline (PBS). Phalloidin 565 was then added as per the manufacturer's suggested dose and incubated

for 30 minutes at room temperature in the dark to probe the actin cytoskeleton. DAPI was used as the counterstain at a concentration of $1\ \mu\text{g mL}^{-1}$ for 5 minutes at room temperature in the dark after being washed with PBS. A fluorescence microscope was used for imaging, using appropriate filter sets for red (Phalloidin 565) and blue (DAPI) fluorescence channels.

2.6.4. Gene expression. The expression of cardiac-specific genes was analysed in rat cardiomyoblast H9C2 cells seeded at a density of 1×10^6 cells per mL in 3D printed M-PVA/GG and M-PVA/GG/rGO hydrogels by quantitative real-time PCR. After five days of culture, the cell-rich hydrogel constructs were treated with the Trizol reagent to extract total RNA. The fold change in mRNA expression was assessed by a $2^{-\Delta\Delta\text{Ct}}$ method. The genes of interest and primer sequences are provided in Table S3.

2.7. Statistical analysis

All data are presented as mean $\pm$ SD. Statistical analysis was performed using SPSS software. One-way ANOVA followed by a *post-hoc* Tukey's HSD test was performed to analyse the significant difference between the groups ($n = 3$; $*p < 0.05$, $**p \leq 0.01$, and $***p \leq 0.001$).

3. Results and discussion

Cardiac tissue-mimetic constructs are crucial for restoring cardiac function after myocardial infarction (MI) through electromechanical integration, structural support, and biological modulation. An electroconductive composite hydrogel, M-PVA/GG/rGO, has been developed to facilitate cardiac repair post-MI. Blending of thermoresponsive gellan gum (GG) with photocrosslinkable M-PVA aims to achieve mechanical integrity for printability and improved strength of 3D structures post-printing. GG in the polymeric blend holds its shape during printing, while M-PVA strengthens the structure post-printing by photo-crosslinking. This combined approach supports the development of stable and well-defined 3D constructs. The incorporation of reduced graphene oxide (rGO) into an interpenetrating polymer network (IPN) formed between photocrosslinkable M-PVA and thermoresponsive GG imparts electrical conductivity, which is essential for re-establishing electromechanical coupling between viable and scarred tissue. The IPN network provides a mechanically stable scaffold capable of withstanding the dynamic load of the cardiac microenvironment. Furthermore, nano- and micro-architectural features were achieved through 3D printing of the formulated ink, allowing for precise control over the construct's structure. The hydrogel matrix was specifically tailored to possess tunable mechanical and electrical properties suitable for cardiac function restoration. Collectively, this construct is hypothesized to restore cardiac function and promote tissue regeneration following myocardial infarction.

3.1. Synthesis and characterization of rGO

Graphene-based biomaterials have been used for a variety of tissue engineering applications owing to their high surface

area, high thermal and chemical durability, outstanding electromechanical characteristics, and physicochemical features. The high concentration of oxygen-containing groups (such as carboxyl, hydroxyl, and epoxy functional groups) in GO nanosheets supports their capacity to disperse in aqueous solutions.^{43,44} The electrically insulating qualities of GO because of this oxygen functionalization restrict its use in the engineering of excitable tissues. To reduce GO, a variety of chemical agents have been employed, including sodium borohydride, hydrazine and ascorbic acid. In this study, a minimum amount of hydrazine hydrate was taken to effectively lower oxygen functionality, enhance electrical conductivity, and promote cell viability and proliferation.^{45,46} Fig. 1A shows the reduction process strongly supported by the UV-vis absorption spectra of graphene oxide (GO) and reduced graphene oxide (rGO). GO usually shows a shoulder peak at around 300 nm, which is ascribed to π - π transitions of the oxygen-containing functional groups, and a large absorption peak at about 230 nm, which corresponds to π - π transitions inside its graphitic domains.⁴⁵ As it decreases, this profile undergoes an important change. The π - π peak in rGO undergoes a bathochromic change, or red shift, to about 270 nm, showing the enhanced and restored conjugated carbon network.

The XRD patterns of GO and rGO, analyzed to determine their crystallinity and interlayer *d*-spacing, are shown in SI, Fig. S4D. The patterns are consistent with previous reports on graphene-based nanomaterials. GO exhibited a sharp, intense peak at 10.93° , corresponding to a *d*-spacing of 8.09 Å. This large interlayer distance and the diminished sharpness of peak are attributed to the intercalation of oxygen-containing functional groups (hydroxyl, epoxy, and carboxyl) between graphene layers, effectively disrupting the ordered graphitic structure and decreasing crystallinity. The rGO peak moved to a higher angle of 24.97° after reduction, which led to a much smaller *d*-spacing of 3.56 Å. The upshift and smaller *d*-spacing indicate partial restoration of the sp^2 carbon network and removal of some oxygen functionalities, allowing the layers to restack more closely.³⁷ The broadening of the rGO peak suggests reduced crystallinity, the presence of residual oxygen groups, and a potentially smaller nanosheet size, factors that can hinder perfect restacking and long-range order.

Raman spectroscopy assessed the alterations in structural order and disorder in graphene-based nanomaterials;³⁷ the GO and rGO spectra are presented in Fig. 1C. The G peak, originating from the $\text{E}_{2\text{g}}$ phonon vibration of sp^2 hybridized carbon atoms, was observed at $1598.61\ \text{cm}^{-1}$ for both GO and rGO, indicating the presence of graphitic domains. The appearance of a prominent D peak at $1357.31\ \text{cm}^{-1}$ for GO and $1352.11\ \text{cm}^{-1}$ for rGO, corresponding to the $\text{A}_{1\text{g}}$ breathing mode of defects and sp^3 hybridized carbon atoms, signifies an increase in structural imperfections within the graphite lattice upon oxidation. Notably, the intensity ratio of the D peak to the G peak ($I_{\text{D}}/I_{\text{G}}$) increased from 0.75 in GO to 0.9 in rGO. This rise in the $I_{\text{D}}/I_{\text{G}}$ ratio of rGO suggests an increase in the number of graphene edges, a reduction in the average size of sp^2 domains, and consequently, a higher degree of structural disorder following the reduction process.

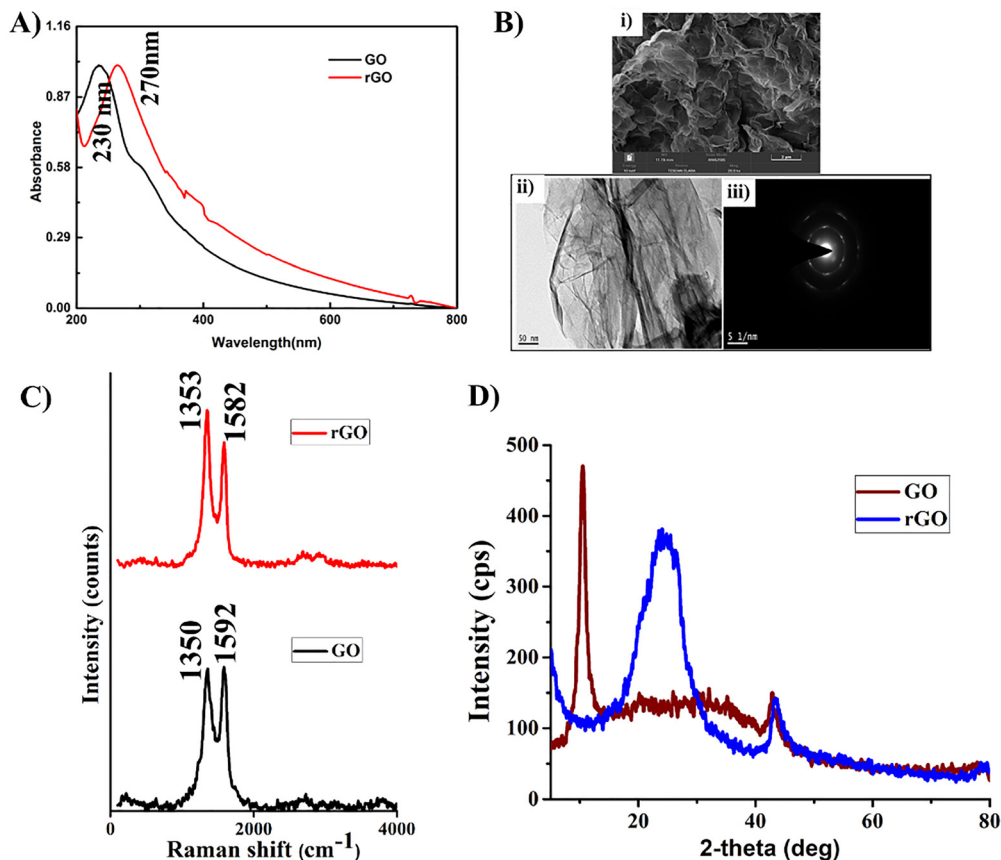


Fig. 1 Characterization of GO and rGO: (A) UV-vis spectra, (B) FE-SEM images of [i] rGO, magnification 20 000 $\times$, [ii] transmission electron micrograph of rGO, magnification 50 000 $\times$ and [iii] selected area electron diffraction pattern (SAED) of rGO, (C) Raman spectra and (D) XRD pattern of GO and rGO.

The morphological observations of GO and rGO were analysed using a SEM, as shown in Fig. 1B. GO nanosheets showed stacked architectures with several crumpled flakes on their surfaces. This shape indicates that graphite nanosheets were deformed throughout the oxidation, restacking, and exfoliation processes. The rGO nanosheets seem to be folded, creating tangled, wrinkled, multilayered patches distinct across edges and ripples. In line with previous reports, these characteristics of rGO samples impart a larger surface area.³⁷ The successful reduction and exfoliation of GO to nanosheets were further observed using TE micrographs. Fig. 1B confirms the nano-size of the particles. In addition, the selected area electron diffraction pattern (SAED) of rGO depicts the clear six point-hexagonal pattern, which is in accordance with previous reports.^{47,48}

The biocompatibility of rGO at various concentrations was investigated using the endothelial cell line (EA hy926) as shown in Fig. S1 for over 1 and 3 days. Cell viability notably increased by day 3 for doses ranging from 31.25 to 250 $\mu\text{g mL}^{-1}$, comparable to the control group. At greater rGO doses, however, there may be delayed cytotoxic effects as 500 $\mu\text{g mL}^{-1}$ dose demonstrated a noticeable drop in cell viability by day 3. These findings are consistent with previous reports, indicating that rGO, at optimized concentrations, enhances cell adhesion and promotes electrical signaling without inducing significant

cytotoxicity, thereby making it a promising conductive biomaterial for cardiac tissue engineering.^{45,46}

3.2. Characterisation of M-PVA

The FTIR spectrum of synthesized M-PVA in Fig. 2B showed a broad band at approximately 3300 cm^{-1} , characteristic of O-H stretching vibration of hydroxyl groups. The presence of this band in both pristine PVA and M-PVA indicates that hydroxyl groups remain in the polymer structure after modification with glycidyl methacrylate (GMA). However, the lower intensity of the O-H band in M-PVA compared to PVA suggests partial substitution of hydroxyl groups, likely due to modification with GMA. Furthermore, the appearance of new bands at 2948 cm^{-1} (C-H stretching) and 1700 cm^{-1} (C=O stretching) is characteristic of methacrylate in the M-PVA spectrum, which confirms the successful grafting of GMA onto the PVA backbone.³⁸

This chemical modification was further corroborated by NMR spectroscopy, Fig. 2C, which showed new signals at δ 5.45 ppm and δ 5.69 ppm, attributable to vinyl protons of the methacrylate group, and signals in the range of δ 2.5–3.0 ppm, characteristic of epoxide protons of GMA, thereby clearly confirming successful modification. The degree of substitution was quantitatively determined to be $23 \pm 1\%$ with a yield of 90%.³⁹ This methacrylation introduces photo-crosslinkable sites onto PVA, a crucial step for enhancing its applicability in 3D

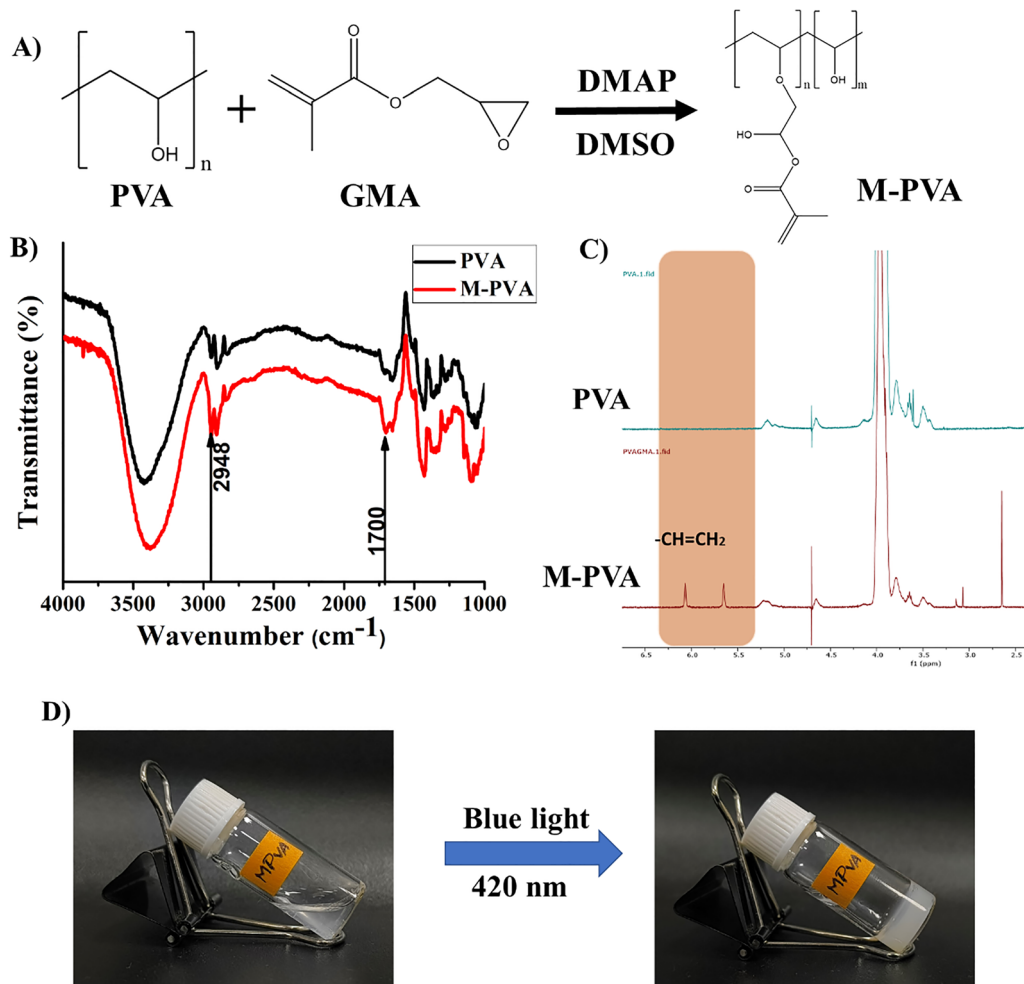


Fig. 2 (A) Synthesis of methacrylated polyvinyl alcohol (M-PVA). Characterization of PVA and M-PVA, (B) FTIR spectra, (C) ^1H NMR spectra, (D) photograph showing M-PVA sol-gel transition and hydrogel formation upon exposure to blue light (420 nm).

printing, particularly for development of photo-crosslinked hydrogels. This modification is anticipated to improve the mechanical properties of the hydrogel and its capacity to promote cell growth and adhesion, thereby making M-PVA suitable for cardiac tissue engineering.⁴⁹

3.3. Fabrication of the hydrogel and characterization

Gellan gum (GG), an anionic polysaccharide, offers a biocompatible method for creating a hydrogel without the use of chemical crosslinkers through a cooling-induced physical gelation process. GG is distributed as single chains in water when heated.⁵⁰ The individual chains undergo a critical conformational shift when the fluid cools, becoming organized double helices (Scheme 1A). The gelation mechanism's first stage is coil-to-helix transformation. Then, these newly created helices join forces to form a continuous, three-dimensional network. Hydrogen bonds and van der Waals forces, which function as physical crosslinks to stabilize the gel structure and trap water molecules, are the main non-covalent interactions that drive this connection. The properties of the resultant hydrogel can be greatly adjusted, depending on variables like the ionic strength

of the surrounding media, the rate at which the solution cools, and the concentration of GG (0.5 to 2%). Table S1 illustrates that the stronger gels were produced by higher GG concentrations, which are suggestive of a denser helical network and more robust inter-helical connections. But these stiff gels required greater extrusion pressures, which made accurate printing difficult. On the other hand, less stiff gels with considerably better printability were created by lower GG concentrations. So, we have chosen 0.5% GG concentration to print along with M-PVA to form a preprinting hydrogel. Combining GG for pre-printing gelation and M-PVA for post-printing crosslinking provides a stable construct (Scheme 1B). The formation of hydrogels using varying concentrations of M-PVA (5 to 10%) and LAP (0.1 to 0.3%) was successfully achieved (Table 1). The images shown in Fig. S5 demonstrate complete gelation within 1 to 5 minutes upon exposure to blue light (420 nm), indicating the efficient photo-crosslinking of methacrylate groups. Based on the viscosity and dispensability, 10% M-PVA was chosen for subsequent blending with thermoresponsive GG (0.5 and 1%), rGO (0.15 mg mL⁻¹) and printing optimization (Table S2). The exposure to blue light initiated photo-crosslinking of M-PVA in

the blend, leading to the formation of a stable interpenetrating network (IPN) hydrogel. This dual crosslinking system (Table S2) leverages the thermoresponsive GG gelation for pre-printing stability and photo-crosslinkability of M-PVA for post-printing structural integrity, thereby providing a robust strategy for fabricating complex hydrogel constructs.

This combined approach allows controlled deposition of the hydrogel at physiological temperature attributed to GG properties, followed by rapid and localized stabilization of the printed structure *via* photocrosslinking M-PVA upon blue light irradiation. The integration of reduced graphene oxide (rGO) into M-PVA/GG/rGO hydrogels promises to improve their mechanical, thermal, electrical conductivity and rheological properties suitable for cardiac tissue engineering. The successful incorporation of rGO into the hydrogel matrix was confirmed by FTIR spectroscopy, which revealed characteristic band shifts indicative of interactions between rGO and the polymer network. As shown in Fig. S2A, comparing the FTIR spectra of M-PVA/GG and M-PVA/GG/rGO, the shifts observed in the O–H (typically 3200–3500 cm^{-1}), C=O (typically 1700–1740 cm^{-1}), and C–O–C (typically 1000–1200 cm^{-1}) bands suggest the successful interaction and integration of rGO into the polymer matrix.⁴¹ The shifts can be attributed to the formation of hydrogen bonds or other non-covalent interactions between oxygen-containing functional groups on rGO and the hydroxyl and carbonyl groups of M-PVA and GG polymer chains respectively.⁵¹ The appearance or enhancement of rGO-specific bands, related to the graphitic structure (typically around 1580 cm^{-1} for C=C stretching in rGO), in the M-PVA/GG/rGO spectrum further confirms its presence and integration within the hydrogel.⁵² The interactions are crucial for the uniform dispersion and effective reinforcement of the hydrogel matrix by rGO.

Differential scanning calorimetry (DSC) evaluated the thermal behaviour of hydrogels. Fig. S2B shows an increase in glass transition temperature (T_g) of M-PVA/GG/rGO (107 °C) compared to M-PVA/GG (91 °C), signifying increased thermal stability upon rGO incorporation. This increase in T_g suggests that the presence of rGO restricts the mobility of the polymer chains within the hydrogel matrix, likely due to the interfacial interactions between rGO nanosheets and the M-PVA/GG network, a phenomenon observed in other nanocomposite hydrogels where the inclusion of nanofillers increases thermal stability by limiting polymer chain movement.⁵²

Scanning electron micrographs in Fig. 3A and B revealed that the dual crosslinked hydrogel has a slight difference in the microstructure, with M-PVA/GG/rGO exhibiting a more uniform and highly interconnected porous architecture compared to M-PVA/GG. The pores were analyzed using SEM images of M-PVA/GG and M-PVA/GG/rGO scaffolds, and pore sizes were measured using ImageJ software. The average pore sizes in M-PVA/GG and M-PVA/GG/rGO scaffolds were $4.1 \pm 1.0 \mu\text{m}$ and $5.7 \pm 1.3 \mu\text{m}$, respectively. In cardiac tissue engineering, 4–5 μm pores facilitate infiltration of cardiac cells while supporting smooth muscle cell activation and deposition of immature collagen, which can gradually remodel into a mature extracellular matrix.⁵³ This improved structural organization

due to rGO integration is crucial for facilitating essential processes in tissue engineering, such as efficient waste removal, oxygen diffusion, and enhanced cell infiltration. Developing a microenvironment that mimics the extracellular matrix (ECM) of native cardiac tissue promotes favourable cell adhesion, proliferation, and alignment, as highlighted in studies emphasizing the importance of scaffold microarchitecture for cardiac tissue regeneration.⁵²

Swelling behaviour is a critical factor in the design of hydrogels for tissue engineering, as it influences hydration, cell viability, nutrient diffusion, and material integrity. The swelling ratio of M-PVA/GG/rGO was notably higher ($\sim 500\%$) compared to M-PVA/GG ($\sim 400\%$), indicating that the addition of rGO improved the ability of hydrogels to absorb water (Fig. S2C). This enhanced swelling capacity supports the creation of a hydrated, cell-friendly environment, essential for maintaining cell function and promoting tissue regeneration.

The degradation profile of hydrogels is another key factor in tissue engineering, as it determines the potential of the scaffold to support tissue regeneration over time. Fig. S2D shows that both M-PVA/GG and M-PVA/GG/rGO hydrogels underwent gradual degradation over 28 days. The M-PVA/GG hydrogel exhibited slightly faster degradation ($\sim 31\%$ weight loss) compared to the M-PVA/GG/rGO composite ($\sim 28\%$). This slower degradation of the rGO-containing hydrogel indicates that inclusion of rGO enhances the structural integrity of the hydrogel matrix, likely due to physical interactions between rGO nanosheets and polymer chains. A controlled and slower degradation rate is desirable in cardiac tissue applications, where scaffold longevity is necessary to support cell growth and matrix deposition during the regeneration process. These results are consistent with previous findings where rGO reinforced hydrogel systems showed improved degradation resistance suitable for tissue engineering.¹⁰

High porosity is essential for cardiac scaffolds to allow cell migration, vascularization, and diffusion of nutrients and oxygen. The porosity analysis reveals that (Fig. S2E) both M-PVA/GG and M-PVA/GG/rGO hydrogels maintain high porosity values ($>90\%$). Specifically, M-PVA/GG showed around $94 \pm 1\%$ porosity, while the addition of rGO slightly reduced the porosity to approximately $91 \pm 1\%$. This minimal decrease is likely due to the compacting effect of rGO nanosheets within the hydrogel network, potentially creating a denser structure. Despite the slight reduction, the porosity remains within an optimal range, supporting its suitability for cardiac tissue engineering. Highly porous structures facilitate rapid cellular infiltration and integration with host tissue, which are critical for the functional recovery of the damaged myocardium.¹⁰

3.3.1. Electrical conductivity. The incorporation of electroconductive rGO into the M-PVA/GG dual cross-linked hydrogel enabled the fabrication of a more biomimetic microenvironment by imparting electrical conductivity, a crucial factor for promoting maturation, function, and contractility of cardiomyocytes. The native myocardium exhibits electrical conductivity ranging from 0.05 mS cm^{-1} (transversely) to 1.6 mS cm^{-1} (longitudinally).³ As shown in Fig. 3C, the M-PVA/GG/rGO

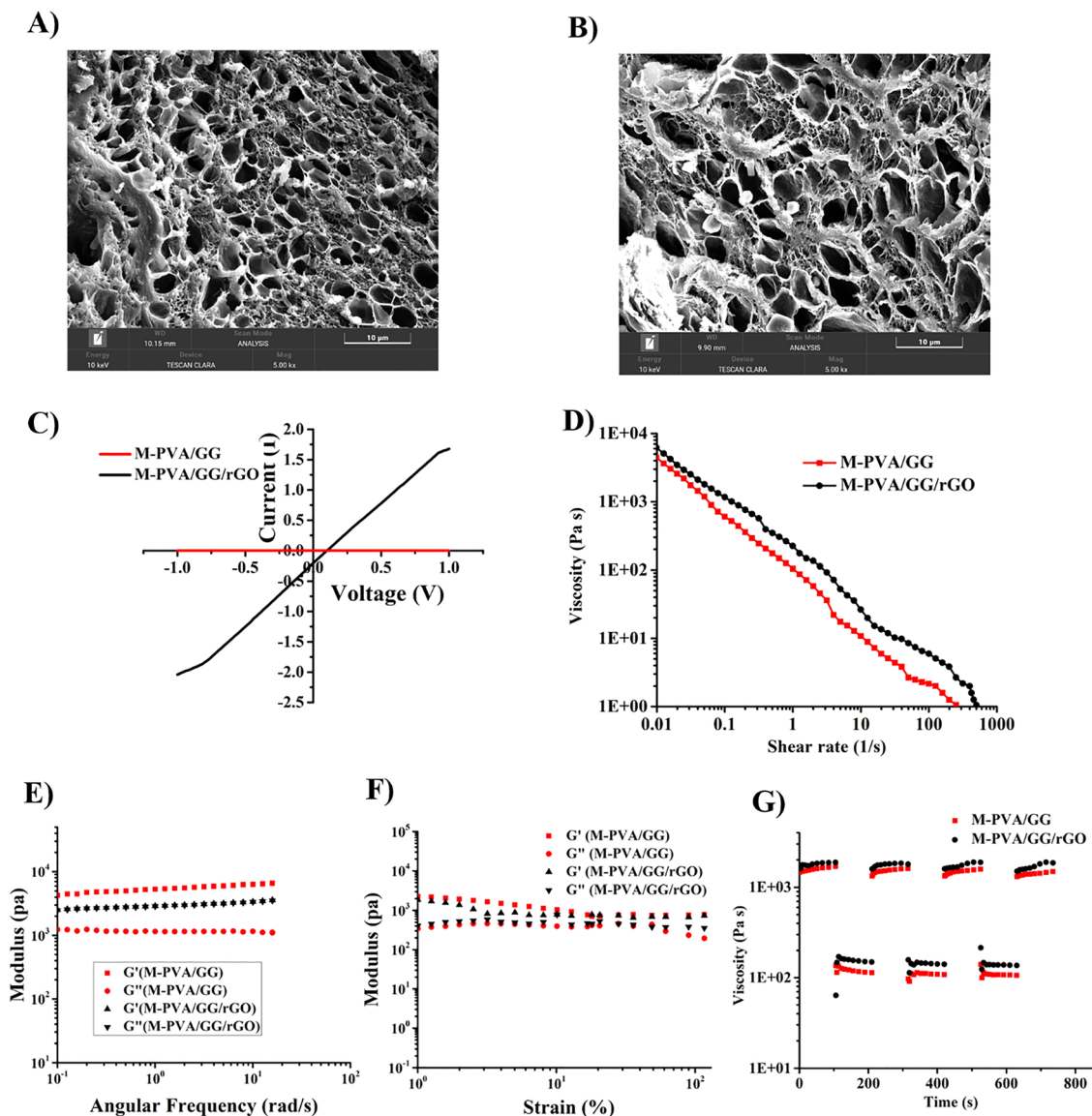


Fig. 3 Characterization of hydrogels, (A) and (B) scanning electron micrographs of freeze-dried M-PVA/GG and M-PVA/GG/rGO hydrogels, magnification 5000 $\times$. (C) electrical conductivity measurements of M-PVA/GG and M-PVA/GG/rGO. Rheological analysis of both hydrogels M-PVA/GG and M-PVA/GG/rGO, (D) viscosity vs. shear rate (E) frequency sweep analysis, at a constant strain of 0.1%. (F) amplitude strain-sweep measurement, at a constant angular frequency (1 rad s⁻¹). (G) Three interval thixotropy analysis, at a constant frequency of 1 rad s⁻¹ and a shear strain of 1%.

hydrogel exhibited a conductivity of 0.62 ± 0.05 mS cm⁻¹, demonstrating an enhancement upon rGO integration. Fig. S3 shows the electrical conductivity test of the M-PVA/GG and M-PVA/GG/rGO hydrogel using a simple LED circuit powered by a 9 V battery. When electrodes were connected to both ends of the hydrogel, the LED lit up, indicating the good electrical conductivity of the M-PVA/GG/rGO composite compared to the control without rGO (M-PVA/GG). The developed M-PVA/GG/rGO hydrogel showed the conductance range of native myocardial tissue, which is essential for effective electrical signal propagation within engineered cardiac constructs.

3.3.2. Rheological properties. The linear viscoelastic region (LVER) (Fig. 3F) determined by amplitude sweep (0.01 to 100% strain) revealed that both M-PVA/GG and M-PVA/GG/rGO

exhibited a stable crosslinked network with a constant storage modulus (G') at low strain. Notably, M-PVA/GG/rGO demonstrated a higher G' within this region compared to M-PVA/GG, which can be attributed to the reinforcement by suggesting a more robust and rigid hydrogel network crucial for providing structural support to developing cardiac tissue.¹⁰ The extended LVER observed in M-PVA/GG/rGO indicates a greater resistance to deformation compared to M-PVA/GG. This enhanced resilience is particularly significant during mechanical stresses encountered in the extrusion-based 3D bioprinting process and for maintaining the structural integrity of engineered cardiac constructs within the dynamic physiological environment of the beating heart. Frequency sweep analysis (Fig. 3E) further confirmed the viscoelastic solid nature of both

hydrogels, with G' consistently higher than the G'' across the tested frequency range. The higher G' exhibited by M-PVA/GG/rGO underscores the increased elasticity and mechanical strength imparted by rGO reinforcement, which is vital for the mechanical properties of cardiac constructs and withstanding cyclic strain.⁹ The frequency-independent G' observed in M-PVA/GG/rGO over a broad frequency range suggests a stable and persistent crosslinked network, a characteristic essential for long-term structural integrity in a dynamically contracting environment of cardiac tissue. Furthermore, the lower G'' maintained by M-PVA/GG/rGO indicates a superior ability to store energy elastically, which could contribute to the efficient transmission of mechanical signals within the engineered cardiac patch. The viscosity vs. shear rate plot revealed that both M-PVA/GG and M-PVA/GG/rGO hydrogels exhibit shear-thinning behavior. The graph shown in Fig. 3D indicates that the viscosity of both formulations lowers as the shear rate increases from 1 to 1000 s^{-1} , demonstrating a shear-thinning tendency that is appropriate for extrusion printing. For both hydrogels, the power law index n was found to be <1 , indicating non-Newtonian shear-thinning behavior. The hydrogel's shear-thinning properties were further improved by the addition of rGO, which is beneficial for smooth extrusion and shape preservation post printing. The shear-thinning behavior of M-PVA/GG and M-PVA/GG/rGO, which is a critical property for extrusion-based bioprinting, enables the hydrogels to flow under pressure and then rapidly recover their viscosity to maintain the structural integrity of the printed construct. The M-PVA/GG/rGO hydrogel consistently display a higher initial viscosity compared to M-PVA/GG. The graph shown in Fig. 3G indicates the three interval thixotropy study, indicating that the hydrogels exhibited reversible shear-thinning behavior. Viscosity reduced dramatically upon high shear disruption and recovered quickly under low shear. Higher initial viscosity and faster recovery were exhibited by M-PVA/GG/rGO compared to M-PVA/

GG. These results indicate improved structural repair and stability, which is advantageous for extrusion-based bioprinting and tissue engineering applications. This elevated viscosity, attributed to the reinforcement provided by the incorporated rGO nanosheets, suggests enhanced structural integrity and mechanical strength of the ink, significant for maintaining shape fidelity of printed cardiac patches and providing initial mechanical support to cultured cardiomyocytes.⁹

3.3.3. 3D printed photo-crosslinking hydrogel. The printability of various polymeric blend hydrogels was assessed using a pneumatics-based 3D bioprinter (NBIL-TRIVIMA, Bangalore, India) with a 22-gauge size needle and 3 mL syringes, by varying parameters such as extrusion pressure and printing speed. The hydrogel composition consisting of 5% M-PVA, 0.5% GG, and 0.15 mg mL^{-1} of rGO demonstrated excellent printability in a pressure range of 0.7 to 0.8 bar and at a printing speed of 200 mm s^{-1} , enabling the fabrication of different constructs. Fig. 4A and D shows the STL file of the printed construct. Fig. 4B shows the printing of a thermoreversible gel ink immediately after being extruded from the 3D bioprinter nozzle. At this stage, the gellan gum component of the ink has formed a temporary gel due to the temperature change. This thermoreversible gel provides initial shape fidelity, allowing the structure to hold its form during the printing process. In Fig. 4C, photocrosslinking has stabilized the hydrogel structure post-printing process. This image depicts the same printed structure after being exposed to a light source (blue light) in the presence of a photoinitiator. The light exposure triggers the crosslinking of the methacrylate-modified polyvinyl alcohol (M-PVA) component of the ink. The photocrosslinking forms a permanent, covalent network within the hydrogel, significantly enhancing its mechanical strength and stability. The structure now retains its intended shape. Fig. 4E and F shows the preprinted structure of the CSIR logo and grid collapsing the structure in the absence of photocrosslinking. These images

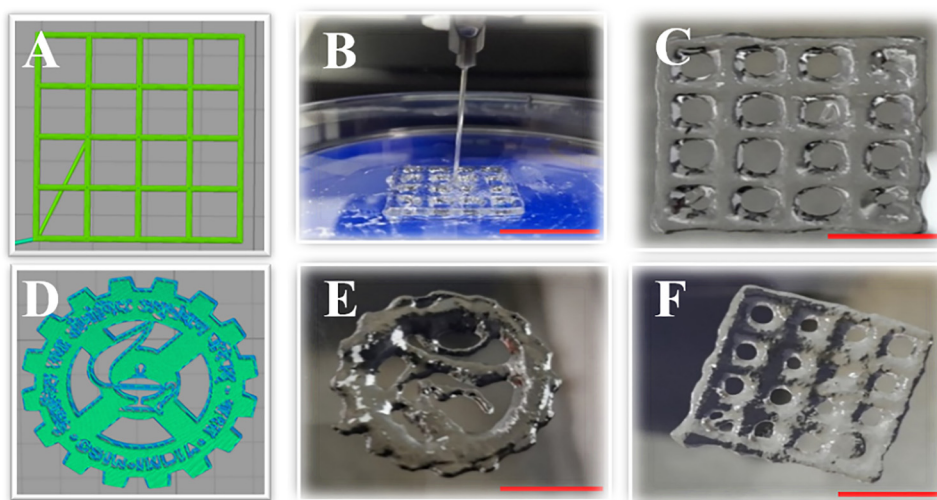


Fig. 4 3D printing optimization of a dual crosslinked M-PVA/GG/rGO hydrogel: (A) and (D) STL file of the constructs. (B) Pre-printing – thermoreversible gel. (C) Stabilization of structures by post printing the photocrosslinking gel. (E) Logo of our council (CSIR) and (F) printed structures collapsing in the absence of photocrosslinking. Scale bar = 18 mm.

illustrate what happens when the printed structures are not subjected to the post-printing photocrosslinking step. The logo (Fig. 4E) and grid pattern (Fig. 4F), with the temporary thermo-reversible gel of gellan gum holding them together, lose their defined shapes and collapse. Fig. S6 shows the printed constructs (B and C), which closely match theoretical design, suggesting that the designed construct (A) was successfully printed with high fidelity. Quantitative measurement (Fig. S6D) verified the printed construct's structural integrity and reproducibility. The printing fidelity results demonstrate that the 3D printed constructs preserve consistent strand spacing and shape, closely resembling the theoretical design. While quantitative analysis reveals no significance difference between theoretical and measured values, macroscopic images validate structural accuracy using Image J software.

Fig. 5 shows the 3D printed dual crosslinked hydrogel, Fig. 5A denotes the CAD models of different shapes and dimensions, Fig. 5B shows 3D printed structures fabricated *via* dual crosslinking and, Fig. 5C shows subsequently freeze-dried 3D printed constructs. The specific formulation exhibited appropriate rheological properties necessary for consistent

extrusion-based 3D printing, allowing for the uniform deposition of multiple layers without clogging or structural collapse. Following the optimization of 3D printing parameters, various geometrically defined structures, including intricate patterns, were successfully fabricated using the 3D bioprinter. This demonstrated the versatility and precision of the developed hydrogel in creating complex constructs suitable for tissue engineering, where precise architectural control is crucial for mimicking native tissue structures. The ability to print complex geometries with this hydrogel composition highlights its potential for creating customized scaffolds with tailored micro-environments for cell growth and tissue regeneration.

3.4. *In vitro* cell studies

3.4.1. Cytocompatibility and viability. The cytocompatibility of the 3D printed construct (Fig. S4B) was evaluated using the EA. hy926 endothelial cell line over three days. Both M-PVA/GG and M-PVA/GG/rGO hydrogels supported high cell viability (>95%) at all time intervals. The inclusion of rGO did not adversely affect cell viability, as it was comparable to the M-PVA/GG hydrogel across days 1, 2, and 3. The excellent bio-

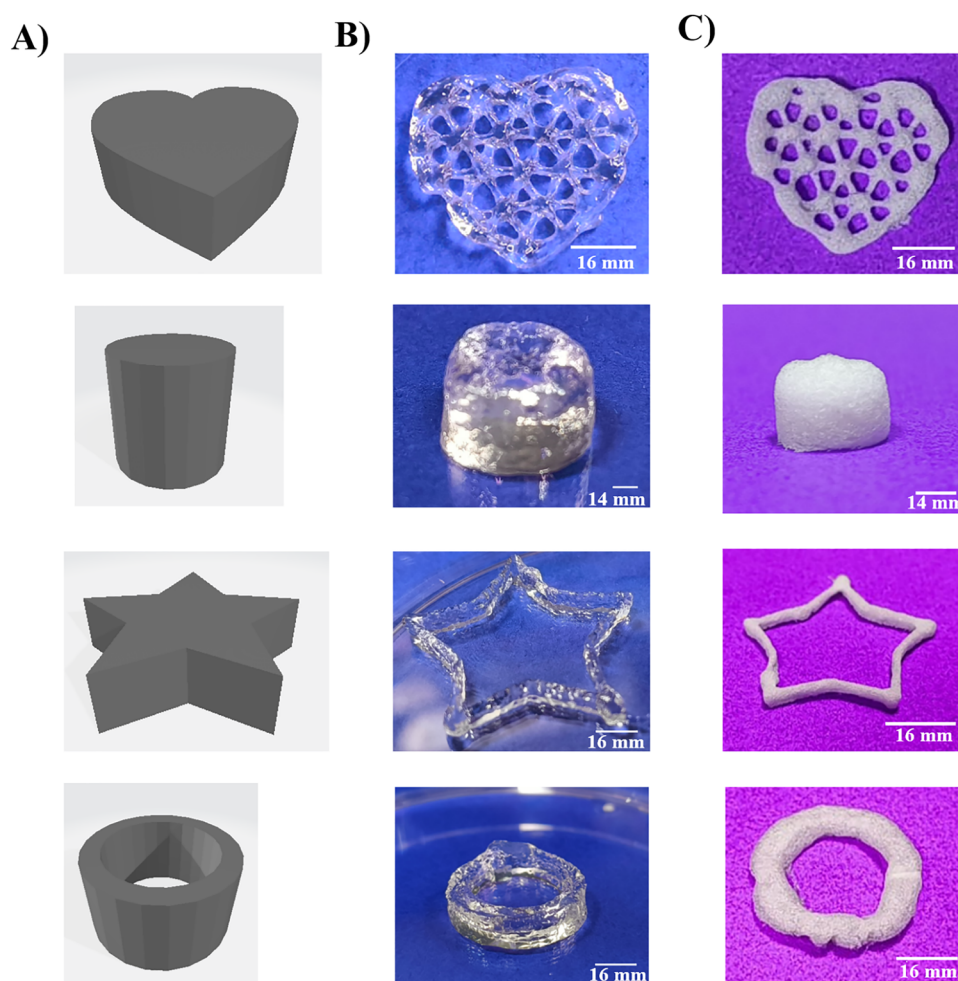


Fig. 5 3D printed dual crosslinked M-PVA/GG/rGO hydrogel: (A) 3D CAD models of different shapes and dimensions, (B) stable 3D printed structures fabricated *via* dual crosslinking, and (C) subsequently freeze-dried 3D printed constructs.

compatibility of the composite hydrogel is crucial for vascularization and endothelialization in cardiac tissue regeneration. Graphene-based materials have been shown to enhance cell-material interactions due to their conductive and bioactive surfaces.¹⁰ Further, the proliferation of H9C2 cardiomyoblasts in a 3D printed construct was assessed over 7 days to determine the hydrogels' ability to support cardiac cell growth. Fig. 6B shows that a significant increase in cell proliferation was observed in the M-PVA/GG/rGO hydrogel compared to M-PVA/GG at days 1, 3, 5, and 7. Particularly at day 7, the rGO-containing hydrogel showed enhanced proliferation, indicating that rGO may provide bioelectrical cues or promote better cell-matrix adhesion. This is consistent with previous studies reporting that electrically conductive nanomaterials, such as rGO, enhance cardiomyocyte behavior, including alignment, coupling, and maturation, favorable for cardiac tissue engineering.⁵⁴ Further, Fig. 6A and Fig. S4A show live/dead cell staining that validated the biocompatibility of hydrogel constructs, showing viable cells (green) with minimal dead cells (red) observed on the printed constructs. These results were consistent with the metabolic assay, further confirming cell viability. To assess cytoskeletal organization (Fig. 7A), phalloidin staining counter-stained with DAPI was performed on 3D printed hydrogels cultured with H9C2 cells, which revealed intact actin filaments. Fig. 6 shows that the hydrogel supports cellular attachment and growth. Overall, the incorporation of rGO into the M-PVA/GG hydrogel enhances its potential as cytocompatible for cardiac tissue engineering.

3.5.2. Gene expression. Fig. 7B presents the influence of rGO incorporation in a dual crosslinked hydrogel on cardiac-specific gene expression assessed by comparing the relative mRNA levels of key cardiac specific markers in M-PVA/GG/rGO and M-PVA/GG hydrogels. A substantial upregulation of all examined genes was evident in the M-PVA/GG/rGO hydrogel.

Specifically, GATA-4 expression, a crucial transcription factor for cardiac development, was significantly enhanced by approximately 15-fold in the M-PVA/GG/rGO group, indicating a pronounced shift towards cardiac lineage. Similarly, alpha-actin, a major component of the cardiomyocyte contractile apparatus, showed an approximately 17-fold increase, suggesting improved formation of the force-generating machinery. The expression of V-cadherin, responsible for cell-cell adhesion, was also elevated by roughly 5-fold, implying enhanced structural integrity. Furthermore, CX-43, a key protein for electrical coupling, exhibited an approximately 3-fold increase, suggesting improved cell-cell communication. Collectively, these results demonstrate that rGO incorporation into the M-PVA/GG hydrogel leads to a significant increase in the expression of key cardiac-specific genes, highlighting its potential to promote a more differentiated and functionally mature cardiac phenotype.^{33,55,56}

4. Conclusion

This study introduces a novel electroconductive, thermoresponsive and photocrosslinkable hydrogel composed of rGO, GG, and M-PVA, designed for 3D printing electro-conductive constructs for cardiac tissue engineering. The printable hydrogel ink formulation helps in the fabrication of 3D constructs that mimic the electrophysiological microenvironment of the native myocardium and leverages the bioactivity of thermoresponsive GG and the mechanical strength of M-PVA. *In vitro* assessments revealed favorable rheological and electrical properties, enhanced proliferation of endothelial cells, suggesting improved vascularization, and increased proliferation and cardiac marker expression in H9c2 cells, indicating promoted cardiac potential. Based on these promising *in vitro* outcomes, the study claims the M-PVA/GG/rGO hydrogel as a strong

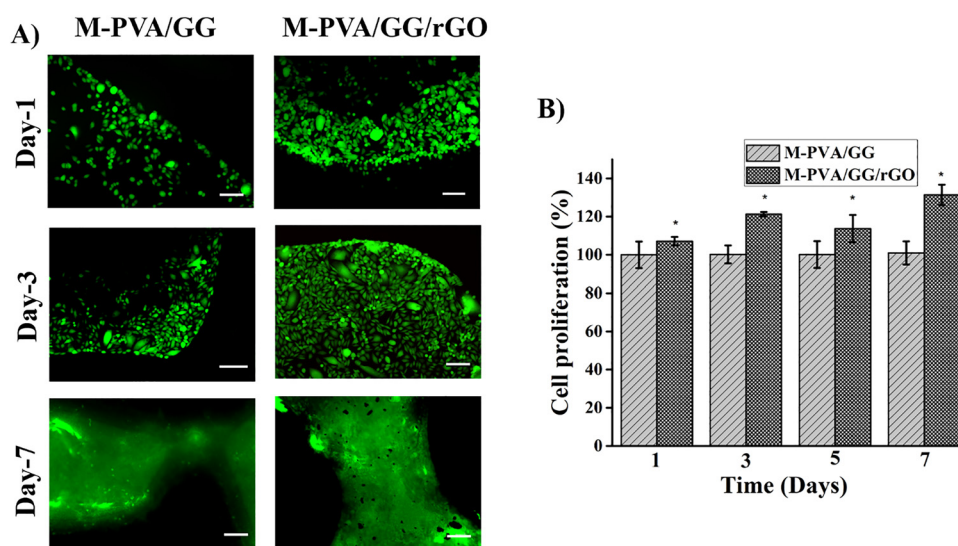


Fig. 6 *In vitro* compatibility of printed hydrogels with and without rGO, using the cardiomyocyte (H9C2) cell line. (A) Fluorescence images showing live/dead staining of cells cultured on the printed hydrogel after 1, 3, and 7 days. Scale bar = 100 μ m; (B) WST assay showing the percentage of cell viability in the printed hydrogel after 1, 3, 5 and 7 days of culture. Statistical analysis was based on one-way ANOVA, Tukey's *post-hoc* test: * $p < 0.05$.

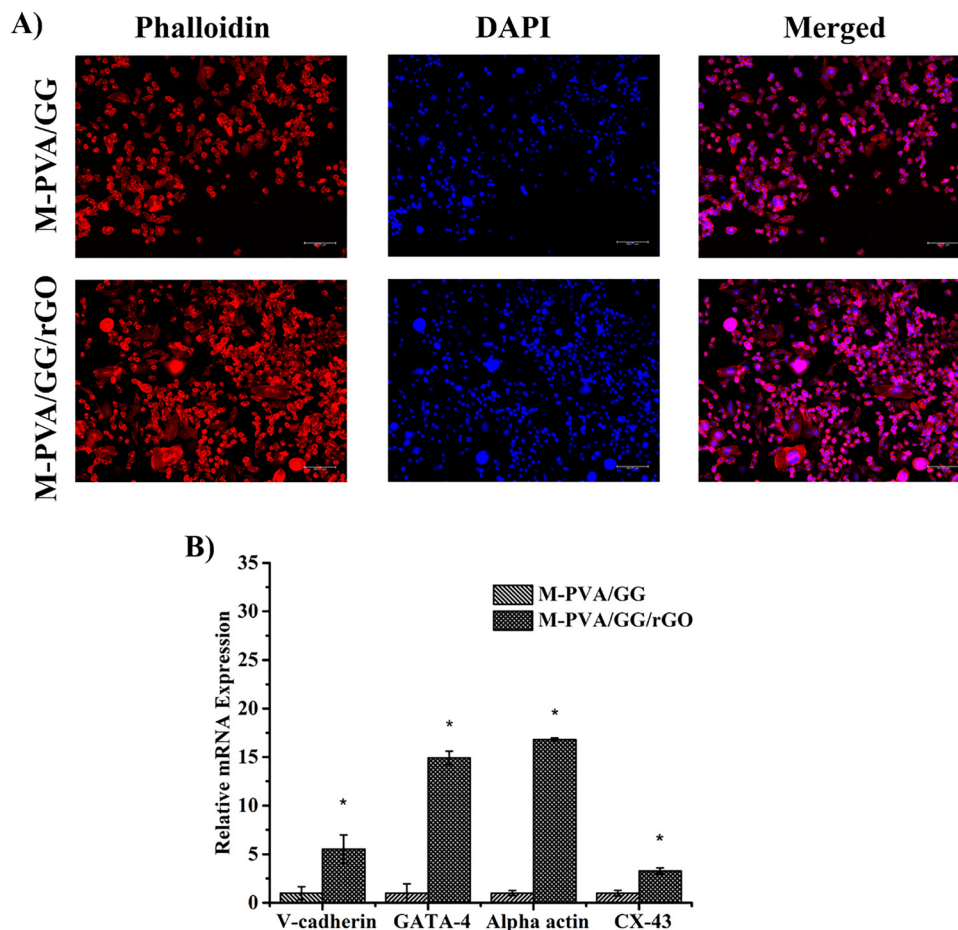


Fig. 7 (A) Cytoskeletal organization of cardiac cells (H9C2) cultured on 3D printed hydrogels, visualized by phalloidin-565 (actin filament-red) and DAPI (Nuclei-blue) staining. (B) Gene expression profiles of Cx-43, GATA-4, V-cadherin, and alpha-actin over a 5-day experimental period for both experiments. Statistical analysis was based on one-way ANOVA, Tukey's post-test: * $p < 0.05$.

candidate for electroconductive cardiac patches and with potential applications in other electrically conductive devices. Future research is essential in animal models of myocardial infarction to evaluate *in vivo* biocompatibility, therapeutic efficacy in cardiac regeneration, vascularization, and electrophysiological integration, paving the way for the development of advanced biomaterials for cardiac repair.

Author contributions

Mohandass Pachaiyappan: writing – review and editing, writing – original draft, visualization, validation, methodology, investigation, formal analysis, data curation, conceptualization; Mercyjayapriya Jebakumar: writing – review and editing, investigation, formal analysis, data curation; Janani Radhakrishnan: writing – review and editing, validation, supervision, conceptualization, formal analysis, data curation, funding acquisition; Niraikulam Ayyadurai: writing – review and editing, validation, supervision, conceptualization, resources, funding acquisition.

Conflicts of interest

The authors declare no conflict of interest.

Data availability

The data supporting this article have been included as part of the SI. Supplementary information: Cell viability of the rGO after 1 and 3 day of culturing (Fig. S1); thermoreversible gel formation of gellan gum (Table S1); optimization of dual cross-linked hydrogel formation (Table S2); characterization of the dual crosslinked hydrogel (Fig. S2); images of electrical conductivity testing of materials using a simple LED circuit (Fig. S3); *in vitro* cytotoxicity test using the printed hydrogel in the EA.hy926 cell line (Fig. S4); optimization of photocrosslinkable hydrogel formation (Fig. S5); printing fidelity of the 3D printed construct (Fig. S6); list of primer sequences (Table S3). See DOI: <https://doi.org/10.1039/d5tb01462j>.

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